EXPLICIT DEFINING RELATIONS FOR THE LEVEL-ONE DEMAZURE MODULE OF A FINITE-DIMENSIONAL COMPLEX SIMPLE LIE ALGEBRA

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Allowed standard prerequisites (editorial).

- Root systems and structure theory of finite-dimensional complex simple Lie algebras. Roots, coroots, the highest root theta, the invariant form normalized on long roots and d_alpha = 2/(alpha|alpha) are assumed. The dichotomy (theta|alpha) in {0,1} for alpha in R^+ different from theta, its consequence for <k theta, alpha^vee>, and the short/long root alternatives are part of the retained proof.
- Local Weyl modules for the current algebra g[t] and their defining relations. The excerpt carries the author's definition of W(lambda), its grading, and the relation (w2') that follows from Garland's identity. Comparing those relations with the defining families of the level-one Demazure quotient is the retained proof.

Lemma 1. *Given $s \in \mathbb{N}$, $r \in \mathbb{Z}_{\geq 0}$ and $\alpha \in R^+$, we have*

$$(x_\alpha^+ \otimes t)^{(s)}(x_\alpha^- \otimes 1)^{(s+r)} - (-1)^s \mathbf{x}_\alpha^-(r,s) \in \mathbf{U}(\mathfrak{g}[t])\mathfrak{n}^+[t] \bigoplus \mathbf{U}(\mathfrak{n}^-[t])\mathfrak{h}[t]_+.$$

Definition 1. Given $\lambda \in P^+$, the local Weyl module $W(\lambda)$ is the cyclic $\mathfrak{g}[t]$-module generated by an element w_λ, with following defining relations:

$$\mathfrak{n}^+[t]w_\lambda = 0, \qquad (h \otimes t^s)w_\lambda = \langle \lambda, h\rangle \delta_{s,0} w_\lambda = 0, \qquad s \geq 0, \qquad h \in \mathfrak{h},$$

$$(x_\alpha^- \otimes 1)^{\langle \lambda, \alpha^\vee\rangle + 1} w_\lambda = 0, \qquad \alpha \in R^+. \tag{1}$$

We note that the relation (1) implies

$$\big(x_\alpha^- \otimes t^{\langle \lambda, \alpha^\vee\rangle}\big) w_\lambda = 0, \qquad \alpha \in R^+, \tag{2}$$

which is easy to see from Lemma 1.

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Definition 2. (see [4, Corollary 3.5]) The level one Demazure module $D(1,\lambda)$ is the graded quotient of $W(\lambda)$ by the submodule generated by the union of the following two sets:

$$\big\{(x_\alpha^- \otimes t^{s_\alpha})w_\lambda : \alpha \in R^+ \text{ such that } d_\alpha > 1\big\}, \tag{3}$$

$$\big\{\big(x_\alpha^- \otimes t^{s_\alpha - 1}\big)^2 w_\lambda : \alpha \in R^+ \text{ such that } d_\alpha = 3 \text{ and } m_\alpha = 1\big\}. \tag{4}$$

In particular, for $\mathfrak{g}$ simply laced, $D(1,\lambda) \cong_{\mathfrak{g}[t]} W(\lambda)$. We denote by $\overline{w}_\lambda$, the image of w_λ in $D(1,\lambda)$.

Proposition 1. *Given $k \geq 1$, the level 1 Demazure module $D(1,k\theta)$ is the graded $\mathfrak{g}[t]$-module generated by an element $\overline{w}_{k\theta}$, with the following defining relations:*

$$\mathfrak{n}^+[t]\,\overline{w}_{k\theta} = 0, \qquad (h \otimes t^s)\overline{w}_{k\theta} = \langle k\theta, h\rangle \delta_{s,0}\overline{w}_{k\theta}, \qquad s \geq 0, \quad h \in \mathfrak{h},$$

$$(x_\alpha^- \otimes 1)\overline{w}_{k\theta} = 0, \qquad \alpha \in R^+, \quad (\theta|\alpha) = 0,$$

$$(x_\alpha^- \otimes 1)^{kd_\alpha+1}\overline{w}_{k\theta} = 0, \qquad \big(x_\alpha^- \otimes t^k\big)\overline{w}_{k\theta} = 0, \qquad \alpha \in R^+, \quad (\theta|\alpha) = 1,$$

$$(x_\theta^- \otimes 1)^{2k+1}\overline{w}_{k\theta} = 0.$$

Proof. Observe that, from the abstract theory of root systems $(\theta|\alpha) = 0$ or 1, $\forall\, \alpha \in R^+ \setminus \{\theta\}$. This implies that $\langle k\theta, \alpha^\vee\rangle = 0$ or kd_α, $\forall\, \alpha \in R^+ \setminus \{\theta\}$. Hence the relations (4) do not occur in $D(1,k\theta)$ and the relations (3) are

$$\big(x_\alpha^- \otimes t^k\big)\overline{w}_{k\theta} = 0, \qquad \alpha \in R^+, \quad \alpha \text{ short}, \quad (\theta|\alpha) = 1.$$

For a long root $\alpha \in R^+$ with $(\theta|\alpha) = 1$, by (2) it follows that $(x_\alpha^- \otimes t^k)\overline{w}_{k\theta} = 0$. Now the other relations are precisely the defining relations of $W(k\theta)$. This proves Proposition 1. □

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